

- 1- An incompressible flow with free-stream speed U_∞ passes over a thin cambered airfoil of chord c .

The mean camber line $z(x)$ is piece-wise defined:

$$z(x) = \begin{cases} \frac{2h_1}{\ell_1^2} x(\ell_1 - x), & 0 \leq x \leq \ell_1, \\ -\frac{2h_2}{\ell_2^2} (x - \ell_1)(\ell_1 + \ell_2 - x), & \ell_1 < x \leq \ell_1 + \ell_2 (=c), \end{cases}$$

Where:

- $h_1=0.015c$,
 - $h_2=0.010c$,
 - $\ell_1=0.4c$,
 - $\ell_2=0.6c$ (so $\ell_1+\ell_2=c$).
- a- Using thin-airfoil theory, derive expressions for the Fourier coefficients A_0 and A_1 .
- b- From your A_0 and A_1 values, compute:
- b-1) the zero-lift angle of attack $\alpha_{L=0}$ (in degrees),
- b-2) the aerodynamics center pitching-moment coefficient C_{mac} .

- 2- An incompressible, inviscid flow of free-stream speed U_∞ acts on a very thin airfoil of chord c . The mean camber line is with a maximum camber $f=0.02c$ at the mid-chord.

$$z_{par}(x) = \frac{4f}{c^2} x(c - x)$$

- a- Using thin-airfoil theory, derive expressions for the Fourier coefficients A_0 and A_1 .
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- b-1) the zero-lift angle of attack $\alpha_{L=0}$ (in degrees),
- b-2) the aerodynamics center pitching-moment coefficient C_{mac} .